

Financial Derivatives II

Two-party option strategies

Protective Put

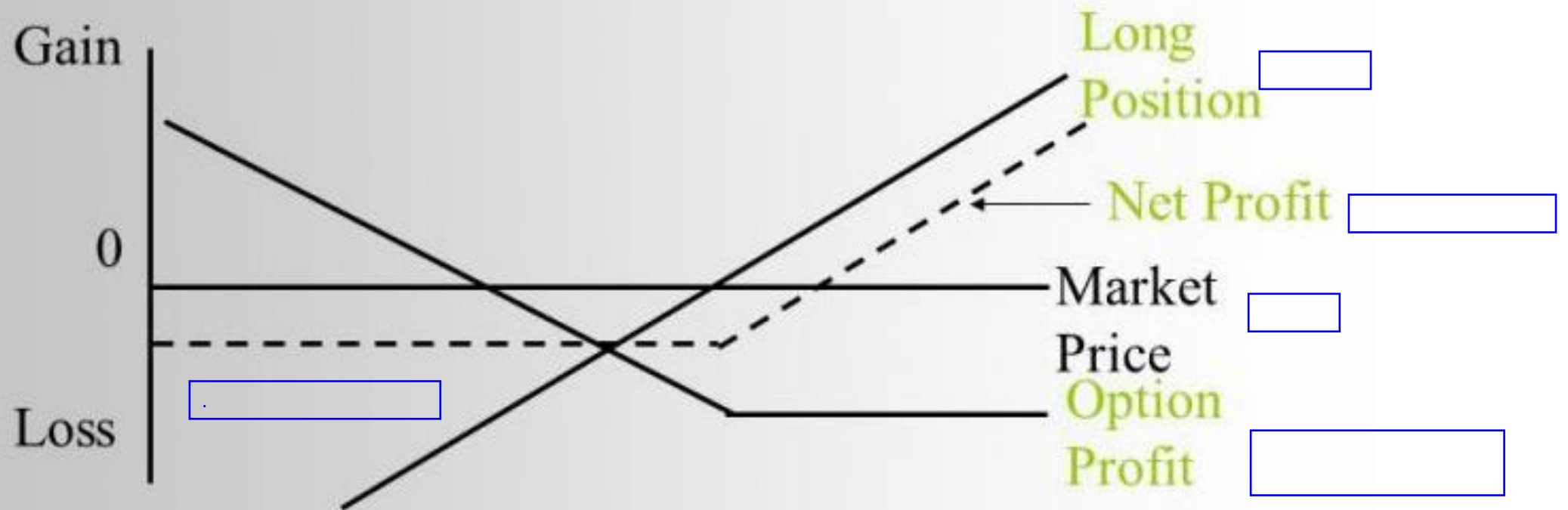
A protective put position is created by buying (or owning) stock and buying put options on a share-for-share basis.

1. To limit risk when first acquiring shares of stock, which is also known as a “married put”
2. To protect a previously-purchased stock when the short-term forecast is bearish but the long-term forecast is bullish

Protective Put

- **Puts** can be used as **insurance against stock price declines**.
- Protective puts **lock in a minimum** portfolio value.
- The cost of the insurance is the put option premium.
- Thus, options can be used for risk management, not just for speculation.

Protective Put Strategy



Covered Call

- A covered call, which is also known as a “buy write,” is a two-part strategy in which stock is purchased and calls are sold on a share-for-share basis.
- Covered call is suitable for neutral-to-bullish market conditions. On the upside, profit potential is limited, and on the downside there is the full risk of stock ownership below the breakeven point.

Benefits of Covered calls

1. The premium received from selling a covered call can be kept as income.

2. Selling covered calls can help investors target a selling price for the stock that is above the current price.

3. Another reason some investors sell covered calls is to get a limited amount of downside protection.

Risks of Covered calls

1. The real risk of losing money if the stock price declines below the breakeven point

downside X

2. The opportunity risk of not participating in a large stock price rise

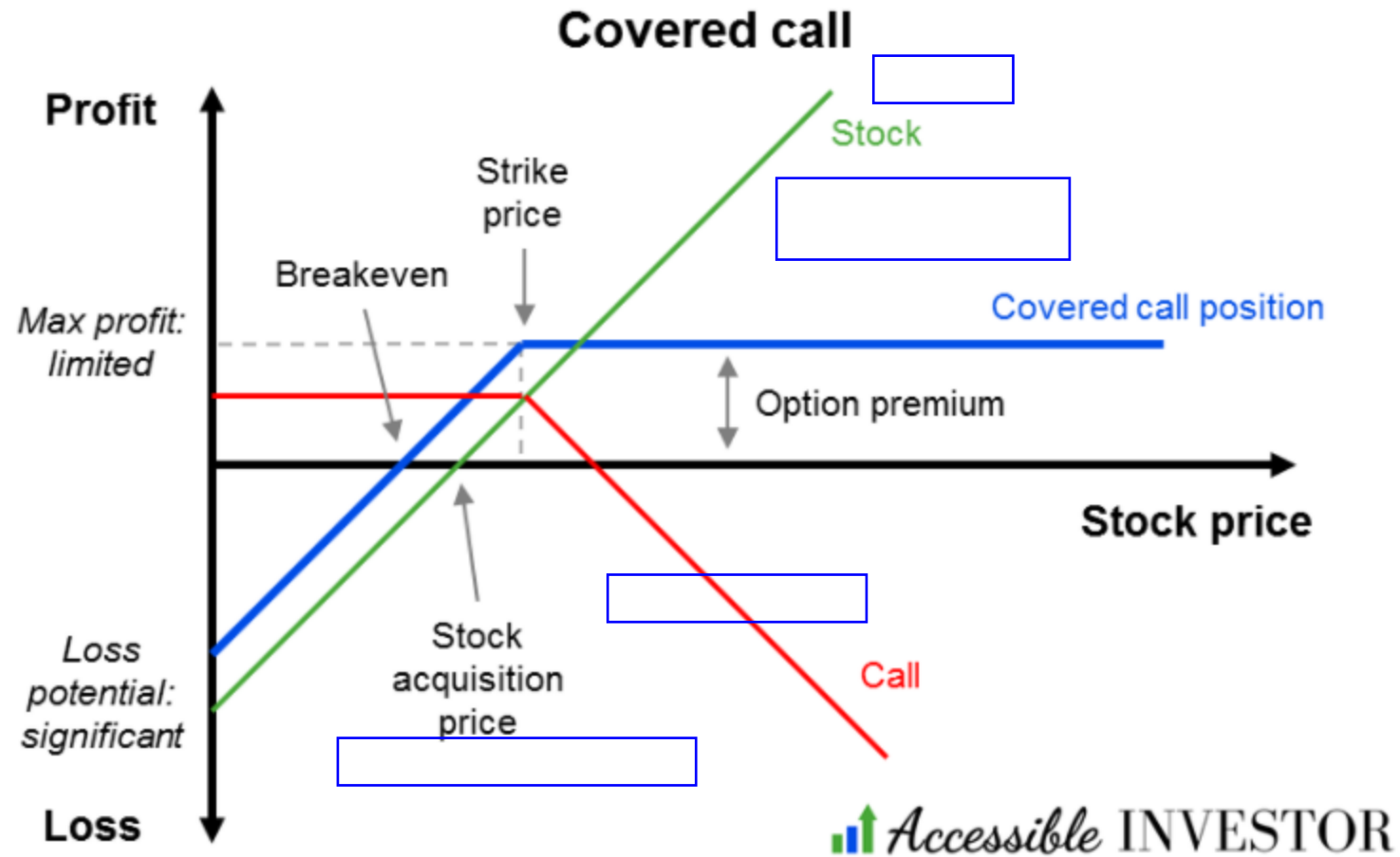
Value of a Covered Call Position at Expiration

Table 20.2

Value of a covered call position at option expiration

	$S_T \leq X$	$S_T > X$
Payoff of stock	S_T	S_T
+ Payoff of written call	-0	$-(S_T - X)$
= TOTAL	S_T	X

Value of a Covered Call Position at Expiration



Straddle

- Long straddle: Buy **call and put with same exercise price and maturity**.
- The straddle is a **bet on volatility**.
 - To make a profit, the change in stock price must exceed the cost of both options.
 - You need a strong change in stock price in either direction.
- The writer of a straddle is betting the stock price will not change much.

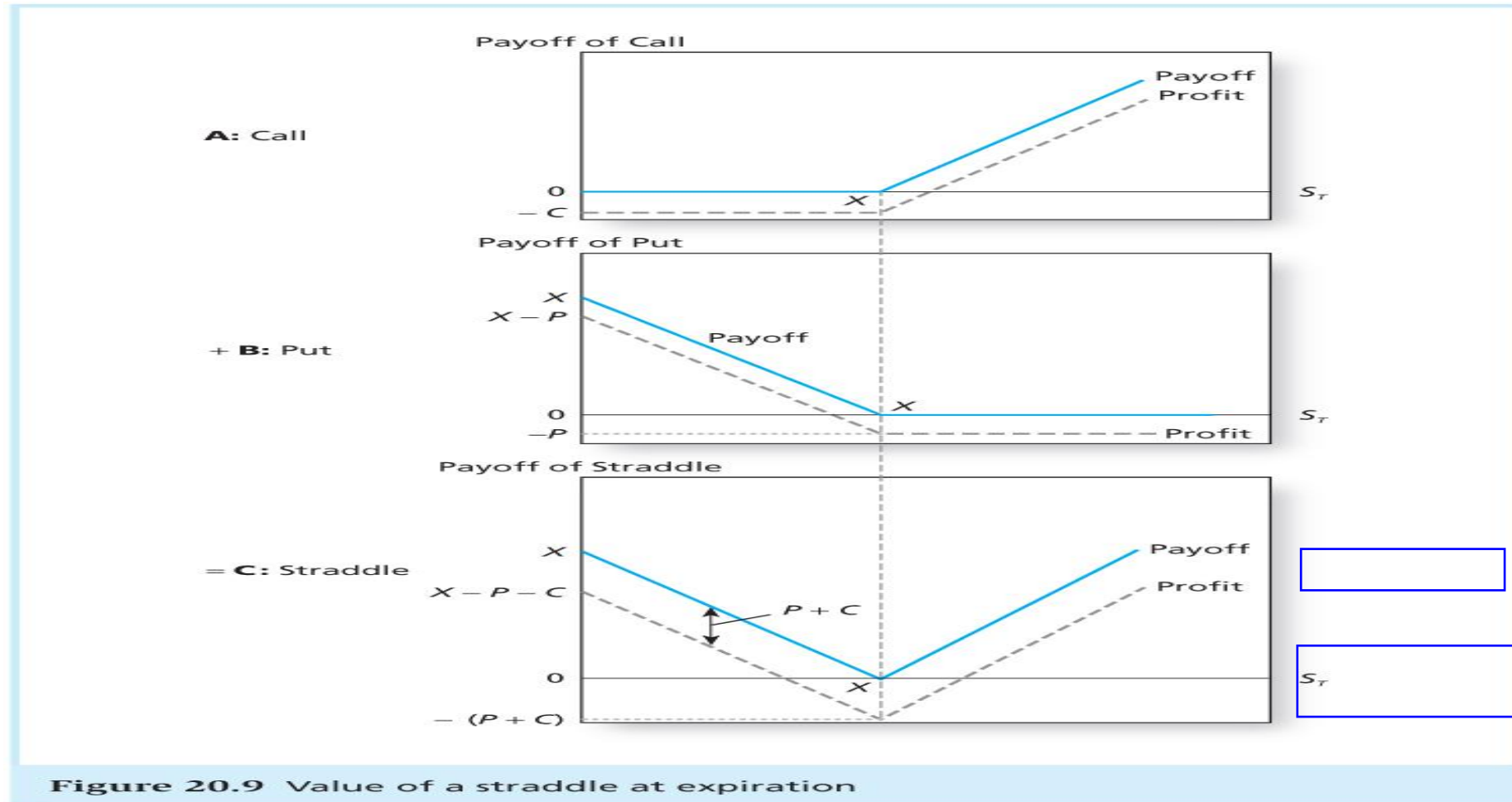
Value of a Straddle Position at Expiration

	$S_T < X$	$S_T \geq X$
Payoff of call	0	$S_T - X$
+ Payoff of put	$X - S_T$	0
= TOTAL	$X - S_T$	$S_T - X$

Table 20.3

Value of a straddle position at option expiration

Value of a Straddle at Expiration



S_T

Spreads

- A spread is a **combination of two or more calls (or two or more puts)** on the same stock with differing exercise prices or times to maturity.



- Some options are **bought**, whereas others are **sold**, or written.
- A bullish spread is a way to profit from stock price increases.

Value of a Bullish Spread Position at Expiration

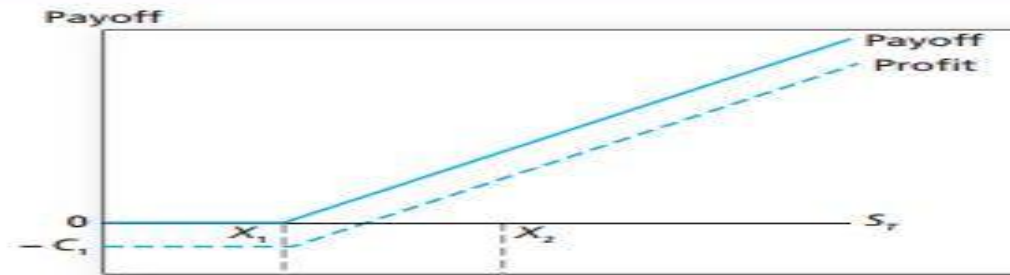
Table 20.4

Value of a bullish spread position at expiration

	$S_T \leq X_1$	$X_1 < S_T \leq X_2$	$S_T \geq X_2$
Payoff of purchased call, exercise price = X_1	0	$S_T - X_1$	$S_T - X_1$
+ Payoff of written call, exercise price = X_2	-0	-0	$-(S_T - X_2)$
= TOTAL	0	$S_T - X_1$	$X_2 - X_1$

Value of a Bullish Spread Position at Expiration

A: Call Held
(Strike price = X_1)



B: Call Written
(Strike price = X_2)



C: Bullish Spread

-> $X_2() X$

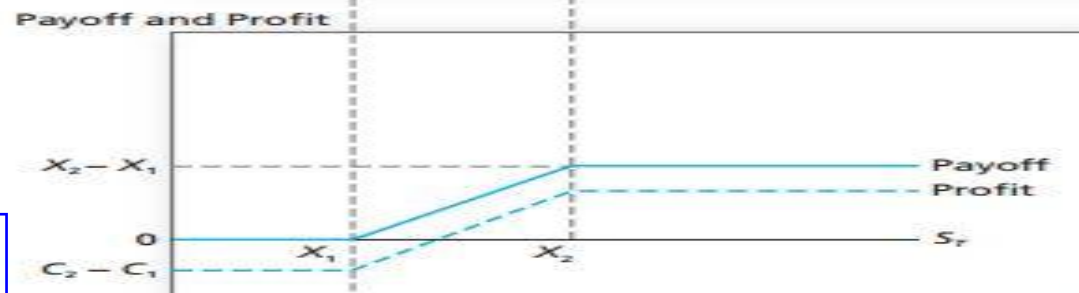
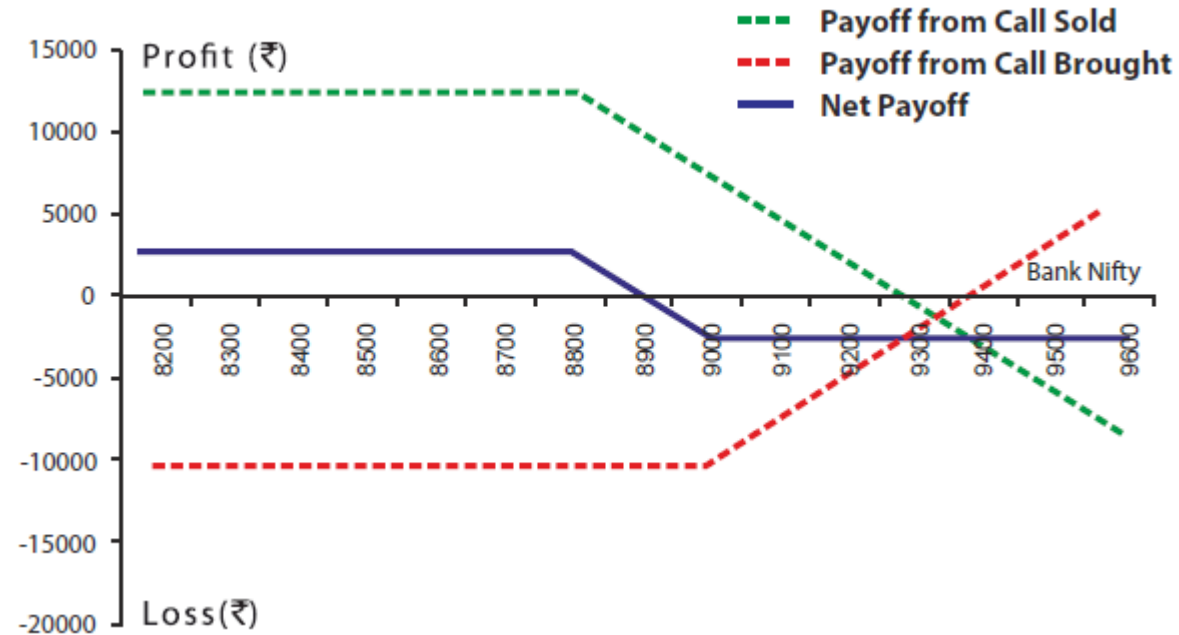


Figure 20.10 Value of a bullish spread position at expiration

Value of a Bearish Spread Position at Expiration



A bear call spread is achieved by simultaneously selling a call option and buying a call option at a higher [strike price](#) but with the same expiration date.

Collars

- A collar is an options strategy that brackets the value of a portfolio between two bounds.



collar position

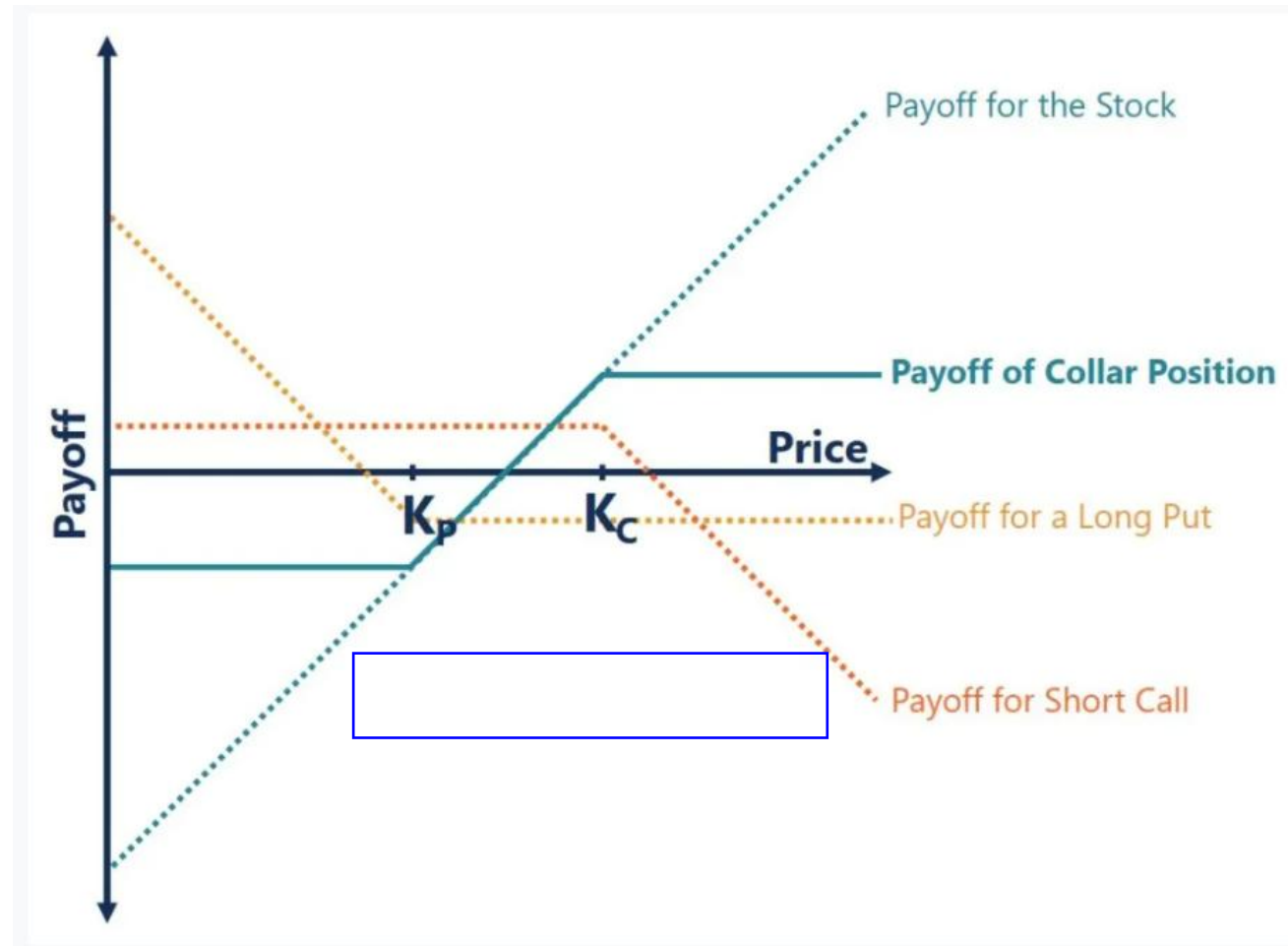


= Long underlying asset + Long put + Short call



- Limit downside risk by selling upside potential.
- Buy a protective put to limit downside risk of a position.
- Fund put purchase by writing a covered call.
 - Net outlay for options is approximately zero.

Value of collar position



Put-Call Parity

- The **call-plus-bond** portfolio (on left) must cost the same as the **stock-plus-put** portfolio (on right):

put-call parity/ ?

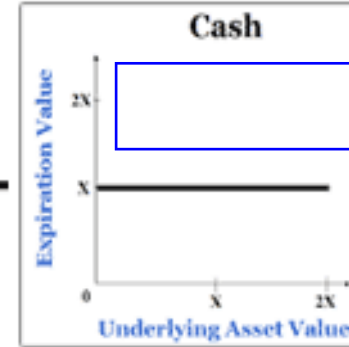
$$C + \frac{X}{(1 + r_f)^T} = S_0 + P$$

Put call parity states that holding up of the long European call with the short European put simultaneously will yield out the same return when you will be holding up a forward contract having the identical basic asset, as well as the expiry date.

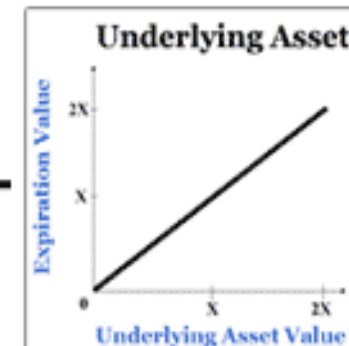
Put-Call Parity

	$S_T \leq X$	$S_T > X$
Value of call option	0	$S_T - X$
Value of zero-coupon bond	X	X
TOTAL	X	S_T

-> face value



	$S_T \leq X$	$S_T > X$
Stock	S_T	S_T
+ Put	$X - S_T$	0
= TOTAL	X	S_T



Put Call Parity - Disequilibrium Example

Stock Price = 110 Call Price = 17

Put Price = 5 Risk Free = 5%

Maturity = 1 yr X = 105

$$C + \frac{X}{(1 + r_f)^T} = S_0 + P$$

117 > 115

overprice(117), un

Since the leveraged equity is less expensive, acquire the low cost alternative and sell the high cost alternative.

Arbitrage Strategy

Position	?	Immediate Cash Flow	Cash Flow in 1 Year	
			$S_T < 105$	$S_T \geq 105$
Buy stock		-110	S_T	S_T
Borrow $\$105/1.05 = \100		+100	-105	-105
Sell call		+17	0	$-(S_T - 105)$
Buy put		-5	$105 - S_T$	0
TOTAL		2	0	0

Table 20.5

Arbitrage
strategy

< 105 0

≥ 105 0